MEDICATION SAFETY SIGNAL MONITOR · HAND-IN R05

Decisions, and what we did not do

Version 1.2, 6 November 2026: the choices that shaped the build, how they changed, and what we left out on purpose

The question this report answers

Which decisions shaped the build, why did we make them, how did they change as the data taught us, and what did we deliberately leave out?

One-line answer

About a dozen decisions carry the design: an explicit formulary, DailyMed as label truth, drug identity resolved once, strict FAERS counting rules, a three-step label check (AI terms, exact text, reviewed synonyms), a cached and grounded AI step, and an incremental pipeline we kept but did not switch on. Several changed version as the data showed us problems. We left out licensed vocabularies, patient data, AI judging signals and anything we could not finish properly.

What this is	What this is not
• An explanation of the decisions that mattered most, each with options, reasons and evidence • A history of how decisions changed (we never edit a decision in place) • A list of what we chose not to do, and why • The approaches we replaced or reverted, kept as part of the record	• The full decision log: that is `docs/06-decisions/DECISIONS.md` (D001 to D048) • A pipeline manual (see the pipeline and Ontology reports) • A list of open decisions: the last 12 proposed ones (D006, D015, D025, D026, D027, D028, D029, D037, D045, D046, D047, D048) were accepted in one batch on 6 November 2026, with D041 v2 • Reviewed by a pharmacist

How to read this. Chapter 1 is the summary and the method. Chapters 2 to 6 each take one theme and explain its decisions: options considered, why, evidence, and status. Chapter 7 is the decision-change history. Chapters 8 and 9 are what we chose not to do and the approaches we replaced. Codes: D### is a decision in the decision log, DQ## a data issue in the data-issues register, A### an approach file (something we tried), O# an entry in the optimization log. Numbers come from `numbers.json`; where a piece of evidence was not captured there, we cite its record (see the appendix table).

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01

Summary and method

The brief says: where it is silent, decide and write it down. We did, in one decision log. A decision has an id, a date, a reason, the alternatives we rejected and a status (Proposed, Accepted or Superseded). A decision is **never edited in place**: when it changes, it gets a new version (D016 v2, v3) or a new id, and a row in a *decision changes* table records the before, the after and the evidence that triggered it. Small realisations go in a *learnings* table.

What the pilot covers. 13 drugs on a hospital formulary: 8 opioids (oxycodone, hydrocodone, morphine, fentanyl, hydromorphone, methadone, buprenorphine, tramadol) and 5 GLP-1 medicines (semaglutide, liraglutide, dulaglutide, tirzepatide, exenatide). Four public sources: FAERS (the FDA's adverse event reports), DailyMed (official label text), openFDA recalls and RxNorm (drug names and IDs). A *signal* is a drug and reaction reported unusually often (PRR, the proportional reporting ratio, with the Evans rule; D006) that is not found in the drug's label.

Status words. *Accepted* = the owner has agreed. *Proposed* = written down and usually built, waiting for the owner's OK. *Superseded* = replaced by a newer version or decision. D039 is the dataset-contracts decision; its gates G1 to G6 settled several earlier decisions during the build (for example G6 accepted D020 to D024 for the pilot).

Twelve decisions were accepted in one batch on 6 November 2026 (the owner's words: "Accept all 12"): D006 (the Evans signal rule), D015 (the AI target, F1 0.85 or more; only the target: its TAC 2017 / OnSIDES benchmark stays cut by D031), D025 (fill-rate monitoring), D026 (the bridge sources Drugs@FDA and NDC Directory), D027 (two-level identity), D028 ('since yesterday' = since last review), D029 (FAERS is quarterly), D037 (the exact-text label check), D045 (change feed), D046 (monitoring), D047 (DailyMed XML reuse) and D048 (no partitioning). All were already built and running. The same day the owner accepted **D041 v2**: the AI also reads the 3 semaglutide labels, and page 1 shows its quotes. D031 v3 (an old schedule) is superseded.

The decisions this report explains

Theme	Decision	In one line	Status
Scope	D018	Formulary = an explicit list of 13 ingredient IDs, not a drug class	Accepted
Sources	D019	DailyMed is the source of truth for labels	Accepted
Identity	D020, D027	Exact ingredient matching; two levels: ingredient → product (→ package)	D020 accepted (via D039 G6); D027 accepted 6 Nov (product key confirmed by D039 G2)
FAERS	D021, D022, D023	Which drugs count; which case version counts; fold duplicates	Accepted
Label check	D041, D037, D043, D016	AI terms with a quote, exact text, then a reviewed synonym table; tag non-harm terms	Accepted (D037 on 6 Nov)
AI	D041, D041 v2	One cached, grounded model call per label section; v2 adds the 3 semaglutide labels	Accepted (v2 on 6 Nov)
Pipelines	D044, D047, D048	Incremental FAERS built and proven, not switched on; DailyMed XML reuse; no partitioning in the pilot	Accepted (D047, D048 on 6 Nov)
Product	D045, D046	Change feed for 'what changed'; monitoring as data	Accepted 6 Nov

The rule behind most choices

Narrow and working beats broad and half-working. Each time we had to choose between breadth and depth, we cut breadth: fewer drugs for AI, fewer sources, a replay instead of live drift.

What this means for Dana (the safety officer)

The core rules are built and accepted. D006 (the signal rule), D037 (the exact-text label check) and D045 (the change feed) run in every build; the owner accepted them on 6 November.

What we left out shapes her screen. No MedDRA licence, so synonyms come from a small reviewed table whose bands no pharmacist has checked yet. FAERS is quarterly, so signals move once a quarter (FAERS is about 130 days behind). The AI reads 5 labels (oxycodone and semaglutide), so for the other drugs 'not in label' comes from the text and synonym checks. For NAION all three checks agree: the AI found diabetic-retinopathy terms and no optic-nerve term.

Every 'no' has a reason and a trigger. Chapter 8 lists what would make us add each missing piece.

Gate: Summary

Required evidence: Each decision below names its options, its reason, its evidence and its status.

Result: Met for the decisions chosen here; the full log runs from D001 to D048.

If it fails: A decision without evidence is an opinion: mark it ASSUMED.

02

Scope and sources

Two early decisions fixed what we watch and which text counts as 'the label'.

D018 · The formulary is an explicit list of ingredient IDs

Part	Detail
Decision	The formulary is 13 RxNorm ingredient IDs (RxCUIs): 8 opioids and 5 GLP-1 medicines (D014 chose this opioid-led mix, A003).
Options	(a) a drug class code (ATC N02A: the opioid group in the WHO's Anatomical Therapeutic Chemical classification); (b) an explicit list of ingredient IDs.
Why (b)	Class codes do not hold our list cleanly: methadone sits in a different class, and hydrocodone appears only through combination codes (DQ31). A list is explicit and auditable.
Evidence	OBSERVED DQ31, seen while profiling RxNorm on 2026-10-20.
Cost	A new drug must be added by hand. That is acceptable for a hospital formulary, which is a list anyway.

D019 · DailyMed is the source of truth for labels

Part	Detail
Decision	Label text comes from DailyMed SPL XML (the official US label library), not from the openFDA copy of labels.
Options	(a) openFDA label API (easy JSON); (b) DailyMed XML (harder to parse, but the original).
Why (b)	The openFDA copy lags: for Ozempic it was months behind DailyMed (DQ28). A stale label gives wrong 'not in label' flags.
Evidence	OBSERVED DQ28 (2026-10-20). We pull 1,321 label versions for our drugs.
Caveats we found later	DailyMed can itself lag the FDA's SrLC database (Safety-related Labeling Changes) (learning, 2026-11-02); and it can answer 'HTTP 200' with the text 'None' instead of XML (DQ42). Raw keeps the reply; clean keeps only real XML.

Gate: Scope and sources TESTED

Required evidence: Each choice is backed by a data issue found while profiling, before building (D012: profile every source by hand first).

Result: Met: DQ31 and DQ28 were found by hand on real data.

If it fails: If a drug the hospital uses is missing, add its RxCUI to the list and rebuild.

03

Drug identity, resolved once

The brief judges whether drug identity is 'resolved once, in the right place'. Ours is resolved in one clean dataset, `clean_drug_identity`, and every other step joins to it.

D020 · Exact ingredient matching, never substrings

Part	Detail
Decision	Split FAERS's active-ingredient text on its separator, strip salt names (hydrochloride, sulfate ...), and match the result exactly to an RxNorm ingredient.
Options	(a) substring or fuzzy name match; (b) exact ingredient after cleaning.
Why (b)	A substring match for 'morphine' also hits apomorphine (a Parkinson's drug) and diamorphine (heroin); 'fentanyl' hits a fentanyl analogue. That inflated morphine rows by about half (DQ34).
Evidence	OBSERVED DQ34, measured on the 2026Q2 FAERS file while profiling on 2026-10-20.

D027 · Two levels: ingredient and product (D003 v1 → v2)

Part	Detail
Decision	Drug = ingredient RxCUI. Product = FDA application number. Package = NDC code. All built once in the identity step.
Before	D003 v1: one join key, the ingredient RxCUI.
Why it changed	Counts belong to ingredients, but labels and recalls belong to products: a semaglutide injection and a semaglutide tablet have different labels, and linking reports to the wrong one created false 'not in label' flags (DQ33). Bulk FAERS has no harmonised label link, but it does carry the application number (DQ39, D024).
Evidence	OBSERVED DQ33 and DQ39. The product key was confirmed as the application number by D039 G2.
Status	Accepted on 6 November 2026 (batch). The two levels are built and used by every label and recall join.

Gate: Identity TESTED

Required evidence: One dataset maps every name we see to an ingredient and a product; nothing downstream matches names on its own.

Result: Met. When identity changes, incremental FAERS replays all quarters as a new generation (D044), so old counts never mix two identity versions.

If it fails: A drug appearing twice under two names would split its counts: check distinct RxCUIs per name after every identity change.

04

Counting FAERS reports honestly

FAERS (the FDA Adverse Event Reporting System) is messy. Three rules decide what one report is, and which drugs in it count.

Decision	Rule	Option rejected	Why	Evidence
D021 suspect roles	Count a drug if it is primary suspect, secondary suspect or interacting, once per report; never concomitant	Primary suspect only	Primary-only would drop more than half of oxycodone cases (learning 2026-11-04); secondary often repeats the primary (DQ35)	OBSERVED
D022 versions and deletes	Keep the latest version of each case; apply deletes to all history; raw quarters stay append-only	Append each quarter	Follow-up versions arrive in later quarters and change the drug list; appending double-counts them (DQ36)	OBSERVED
D023 duplicate fold (v1 → v2)	Fold the same person filed by many companies; v2 also sets aside every report listing 30 or more drugs	Delete those reports	A mass-polypharmacy cluster (one patient re-filed hundreds of times, ages changing) topped the list with unrelated reactions	OBSERVED

How we know

D022, measured in Foundry: the 2026Q2 files replaced 26,681 cases and deleted 655 cases from 2026Q1. That is 6.9% of the 397,224 cases in the 2026Q1 file touched in one quarter DERIVED. Appending quarters would have counted those as new reports.

After all three rules: 792,346 current cases, 373,522 drug-reaction rows for our drugs OBSERVED.

Why D023 v2 keeps the mass reports visible. Deleting them would hide evidence. Instead each signal carries how many of its reports came from mass reports, and a signal that is mostly mass reports gets a caution and is not ranked. A blunt cut, kept visible, is easier to audit than a silent delete DESIGNED.

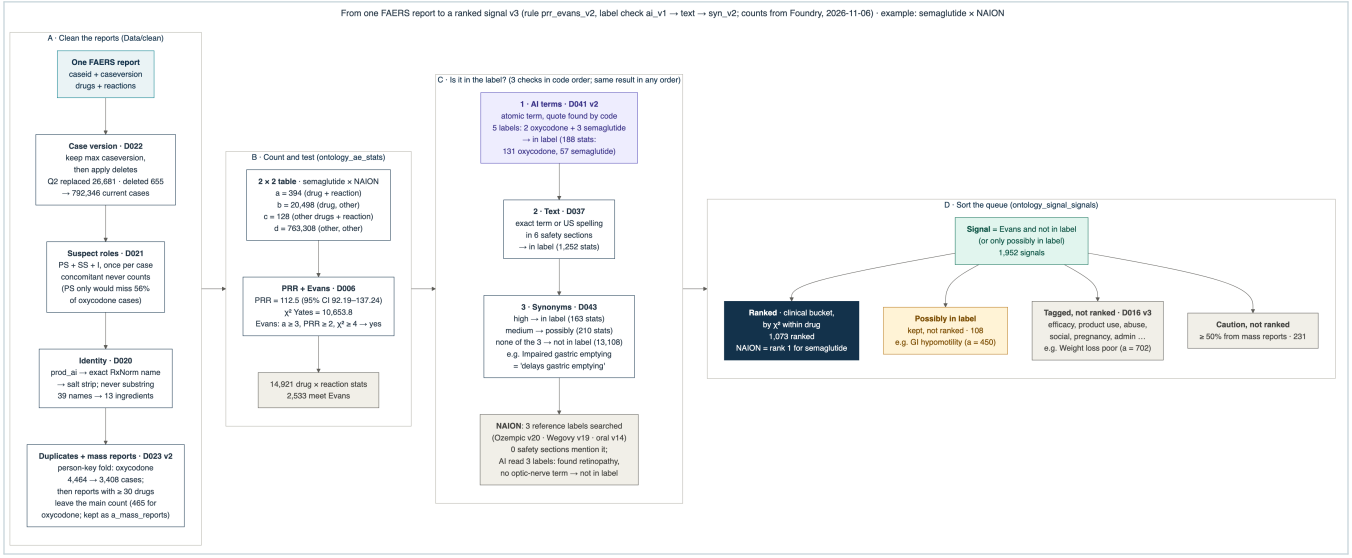


Figure 1. From one FAERS report to a ranked signal: case version (D022), roles (D021), identity (D020), duplicate and mass-report fold (D023 v2), then statistics, the three label checks in code order (AI, text, synonyms) and the buckets.
Source: docs/diagrams/png/32-handin-signal-pipeline-v3.png

Gate: FAERS rules TESTED

Required evidence: Counts equal an independent prototype; the follow-up effect is measured.

Result: Met: Foundry counts equal the 20 October Python prototype exactly (learning 2026-11-04); five random signals re-counted by hand at the baseline gate.

If it fails: If a count moves without new data, a rule changed: check the rule version in the signal key.

05

Is it in the label? Three checks and the buckets

A signal is only interesting if the label does not already mention it. Deciding 'mentioned' is the hardest judgment in the project, and the one that changed most.

Step	Decision	What it does	Outcome
1 · AI terms	D041 v2	Is it among the reactions the AI extracted, with a quote that code found in the label? (the 5 AI-read labels only)	in label / not
2 · Text	D037	Is the exact FAERS term (or its US spelling) in a safety section of the drug's reference label version?	in label / not
3 · Synonyms	D043	Does a reviewed synonym phrase for that term appear in the label?	in label (high) / possibly in label (medium)

The code runs the checks in this order (`ontology_signals.py` l.163-176). The outcome is the same in any order; only the recorded match method (`label_match_method`) differs. First matches over all 14,921 stats (Foundry, after D041 v2): AI 188, text 1,252, synonym high 163, medium 210, none 13,108.

Why a reviewed synonym table and not MedDRA or an LLM judge (D043). MedDRA (the standard medical terminology, whose groupings would solve most synonym cases) needs a licence we do not have for a public pilot. An LLM reading every label for every signal is costly, ungrounded unless it quotes, and would drift between model versions. A small table is cheap, deterministic, versioned and auditable. Synonym version syn_v2 has 258 rows: 69 high, 72 medium and 117 rejected (kept so they are not reviewed twice). The bands are Claude's draft; a pharmacist has not reviewed them ASSUMED.

Why buckets (D016). Without MedDRA's groupings we tag reactions with keyword buckets. Only *clinical* reactions are ranked; efficacy complaints, product-use errors, abuse and overdose, pregnancy, social and legal terms, and administrative terms stay visible with their bucket but are not ranked.

How the ranked queue moved, on identical data

Step	Change	Ranked signals	Evidence
Step 9	First full build: 13-drug formulary, exact text + AI	1,355	OBSERVED
D043 syn_v1	Synonym table v1 + US-spelling fix (10.7% fewer)	1,210	OBSERVED
D016 v3	Wider efficacy and product-use buckets (tagged, not removed)	1,115	OBSERVED
D043 syn_v2	Synonym table v2 (top-12-per-drug review)	1,073	OBSERVED

Today: 1,952 signals, 1,844 not in label and 108 possibly in label; 1,073 ranked ('step' = a stage in our numbered build order). Bucket changes only re-tag, so signal keys and evaluation ids stay the same and saved triage decisions still match (learning 2026-11-05).

Watch out

The ranked queue is still a review queue. Each step removed known false positives (wording misses, efficacy complaints), but nobody has measured how many of the remaining signals a pharmacist would call real.

Gate: Label check TESTED

Required evidence: Each step is versioned (`text+ai_v1+syn_v2`) and its effect measured on the same data.

Result: Met; Foundry equals the local runner for every version (A014).

If it fails: A synonym row that hides a real gap would be a false negative: every row records the phrase, label and section it matched, so it can be reviewed.

06

The AI step, incremental builds and the new pieces

D041 and D041 v2 · AI label extraction: cached and grounded

Part	Detail
Decision	A Python transform calls Claude Sonnet 5.5 (through Foundry AIP) once per label safety section. Each reaction comes with an exact quote, and code checks the quote is in the section text. Results are cached by setid + version + section + text hash + prompt and model version.
Options	AIP Logic function per label; Pipeline Builder's 'Use LLM' block; other models (GPT, Claude Opus).
Why	Per section keeps prompts short and failures local; the quote check catches invented text; the cache means an unchanged label never costs a second call (D005).
v2 (6 Nov)	Same model, prompt and cache; the AI also reads the 3 semaglutide reference labels (Ozempic v20, Wegovy v19, oral Ozempic v14): 5 labels, 21 sections. Trigger: evaluator v2 found the AI and the screen barely met; semaglutide is the demo drug (A019).
Evidence	OBSERVED 745 of 745 quotes found word for word (5 labels). The widening build made 12 model calls (new sections only); a forced rebuild made 0 (O1b). O1: a rebuild went from 9 model calls and 70,776 tokens to 0 and 0. F1 0.97 against the target of 0.85 (D015), on a 35-item answer key (Boxed Warning and Contraindications of the 2 oxycodone labels, written by Claude and frozen before the run, no pharmacist yet).
Honest limits	Only 5 labels and 21 sections. The answer key is small (the AI found 35 of 35 answer-key items), covers only oxycodone, and a pharmacist has not checked it; the semaglutide sections are not scored. The AI does not change the signal list today: text matching already covers its terms on all 5 labels (signals unchanged after v2). Its value is the exact quote, now on the officer's screen (page 1), and an independent 'no optic-nerve term' reading for NAION.

D044 · Incremental FAERS: built, proven equal, not switched on

The brief asks for at least one incremental pipeline. We built one that adds a quarter to a kept state with the same rules as the full rebuild. It produces exactly the same rows (EXCEPT ALL both ways = 0). But at two quarters it is **slower**: 88 s against 32 s for a full rebuild (2.75 times) OBSERVED. Reading and rewriting the state costs more than parsing one quarter. So the full rebuild stays the input to signals until the incremental path is measured faster, probably at four or more quarters ASSUMED.

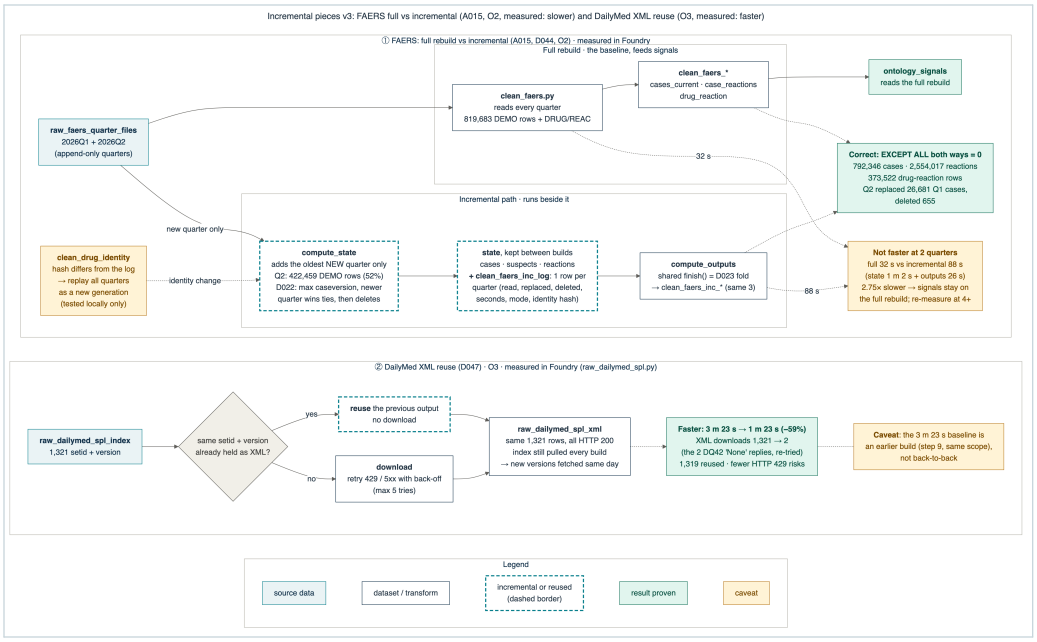


Figure 2. Full and incremental FAERS side by side: state and log, replay when identity changes, EXCEPT ALL = 0.
Source: docs/diagrams/png/33-handin-incremental-faers-v3.png

D047 · Reuse DailyMed XML that has not changed

The slowest build re-downloaded every label XML. Reusing XML for an unchanged setid and version cut downloads from 1,321 to 2 and the build from 3 m 23 s to 1 m 23 s (59% less), with the same rows. The index (about 30 pages) is still pulled every build, so a new label version still arrives the same day **OBSERVED**. Caveat: the 'before' time comes from the step 9 build of the same scope, not a back-to-back run (O3).

D045 · A change feed for 'what changed'

'Since yesterday' means 'since my last review', because sources update on different clocks (DQ14). Many changes are already dated by the source (a recall's report date, a label's effective date); the rest are found by comparing this build with the last. The first build gave 2,898 events; a rebuild on unchanged data added 0 **OBSERVED**. The Morning brief card counts the last 30 days (26); page 3's date range is how the officer narrows it to 'since my last review'.

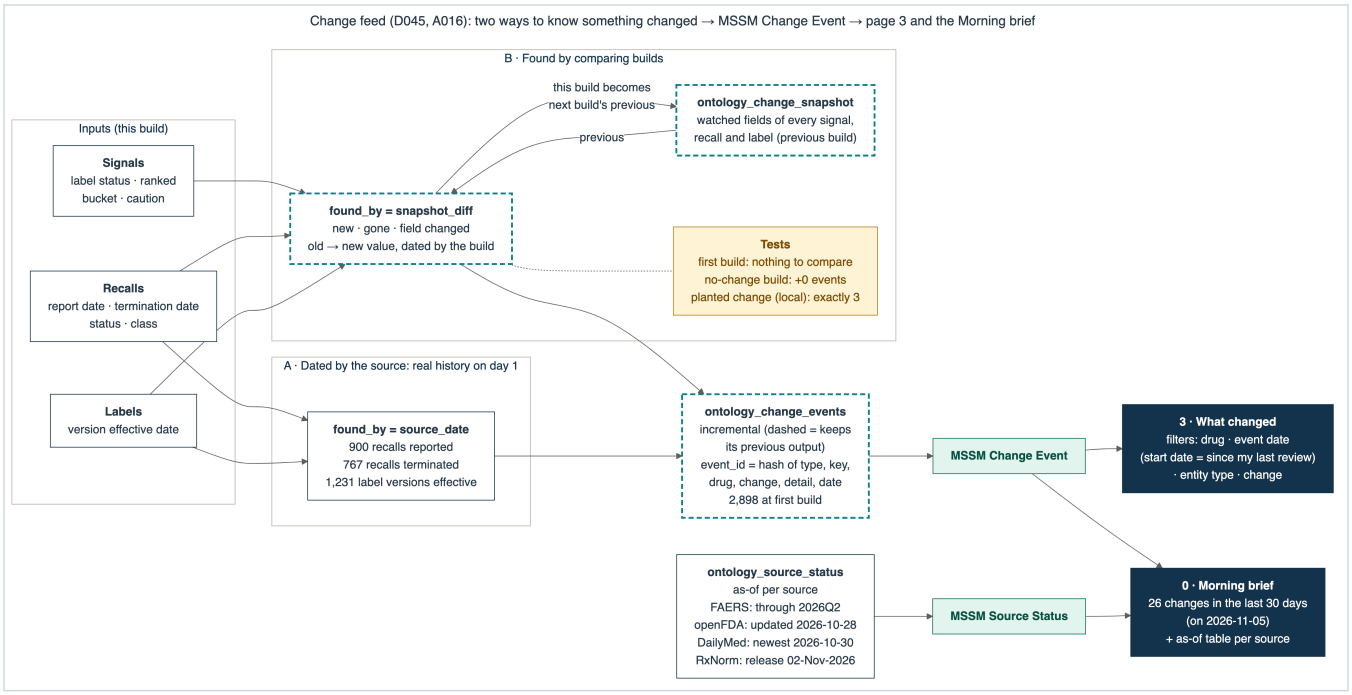


Figure 3. Change feed: source-dated events and build-to-build differences flow into MSSM Change Event and page 3.
Source: docs/diagrams/png/34-handin-change-feed-v1.png

D046 · Monitoring as data

Every build appends 21 checks (keys, row-count drops, freshness, AI grounded rate, cache hits, field fill rates) to a dataset, and rebuilds a drift view as a **historical replay** from source dates, labelled as such. 4 Foundry Data Health checks alert the owner. The first run found two real shifts: FAERS reports with no reporter type rose from 34% to 72% between quarters (DQ38), and recalls for our drugs in the last four weeks were 10 against 3 before OBSERVED.

D048 · No partitioning in the pilot

Every dataset is written unpartitioned. The largest table has 2,554,017 rows, and most build time is most likely container start-up (not measured separately), not data. We revisit (FAERS by quarter, signals by drug) when FAERS holds 8 or more quarters or a single build passes 5 minutes (D048) DESIGNED.

Gate: AI and pipelines TESTED

Required evidence: Every optimization claim has a before and after number, including the ones that failed.

Result: Met: O1 and O3 helped; O2 (incremental) and O3a (column pruning) did not, and both are logged.

If it fails: If incremental is switched on without a new measurement, the claim is unproven: re-measure first.

07

How decisions changed over time

Most changes were triggered by the data, not by opinion. Each row below is a decision that got a new version, with what triggered it.

Decision	Before	After	Trigger (evidence)
D002 → D007 → D014 class	GLP-1 only, then opioids only	8 opioids + 5 GLP-1	Project sponsor's suggestion, then research scoring the hybrid higher (A003)
D003 → D027 identity	One key: ingredient	Ingredient → product → package	Labels and recalls belong to products (DQ33)
D004 v1 → v2 (D022)	Incremental = append each quarter	Upsert latest version + deletes on all history	Phantom rows from appending (DQ36)
D016 v1 → v2	Tag abuse, product use, pregnancy, lawyer-filed	Add social/legal, efficacy, administrative	First ranked list led by child neglect, homicide, imprisonment (A012)
D016 v2 → v3	GLP-1 efficacy and 'Product ...' complaints ranked	Wider efficacy and product-use buckets	GLP-1 lists topped by hunger and 'weight loss poor' (A013); ranked 1,210 → 1,115
D023 v1 → v2	Fold same-person duplicates	Also set aside reports with 30+ drugs	Mass-polypharmacy cluster the person key missed (A012)
D031 v1 → v5 schedule and scope	Demo date, cut list, 11 drugs	No fixed date; 13-drug formulary restored	The schedule slipped a week and the fixed date was dropped; the one-drug slice passed build steps 6 to 8, so gate D039 G1 restored all 13 drugs
D043 data syn_v1 → syn_v2	Synonym table v1	258 rows	Top-12-per-drug review; ranked 1,115 → 1,073
D028 → D045	'Since last review' as an idea (D028)	A built change feed (D045)	Step 11 build (A016)
D041 → D041 v2	AI reads 2 oxycodone labels; 0 for the demo drug	Also the 3 semaglutide labels; page-1 table of AI quotes	Evaluator v2: the AI and the screen barely met (A019); owner, 6 Nov
12 decisions Proposed → Accepted	Built and measured, awaiting the owner	Accepted as built (D015: the target only)	Owner's batch decision, 6 Nov 2026

Note

D031 (schedule) went through five versions: a demo date, a re-ordered cut list tied to hours, a one-week slip that forced all cuts, 'no fixed date', and finally the full 13-drug formulary restored. The cuts that stayed are in chapter 8.

Gate: Change history TESTED

Required evidence: Every changed decision has a dated row with before, after and trigger.

Result: Met in the *Decision changes* table of DECISIONS.md; each change also bumped the Compass version.

If it fails: A decision edited in place loses its history: restore it from git and add a version row.

08

What we chose not to do

A partial build with clear reasoning beats a complete one you cannot explain (the brief). These are the things we left out on purpose, and what would change our mind.

Not done	Why	What it costs us	What would change our mind
MedDRA (licensed medical terminology)	Licence not available for a public pilot	Keyword buckets and a synonym table instead	A licence: replace buckets with MedDRA groupings
Patient-level or private data	Brief: public data only; no PHI	No outcome data, only reports	Never in this project
An LLM judging each signal	Cost; ungrounded unless it quotes; drifts with model versions (D043)	Synonyms need human review	AI proposes synonym rows, a pharmacist accepts them
Live drift alerts	Drift needs 3 to 4 weeks of builds (D009); we had days	Drift is a historical replay, labelled as one	Scheduled builds for a month
Scheduled builds	Pilot built by hand, step by step	Freshness checks do not fire on their own	Production hand-over
AI on all labels	Answer key not reviewed by a pharmacist (D031 v5)	Only 5 labels (oxycodone, semaglutide) have AI quotes	A reviewed gold set on long sections
ClinicalTrials.gov, OnSIDES (open label side-effect data, CC BY 4.0)	D031 cut list (v3 cuts, kept by v5): breadth after depth	No 'seen in trials' or second 'is it labelled?' opinion	Another month (roadmap)
Daily FAERS	No confirmed official daily API (D029). FAERS moved to AEMS (the FDA's new adverse event system) in March 2026 (DQ20); the future feed is unverified	FAERS is quarterly, about 130 days behind	An official AEMS export
SrLC (FDA Safety-related Labeling Changes) backtest	D031 cut list (v3 cuts, kept by v5)	Cannot show 'we would have flagged it first'	Another month
Medicare Part D prescribing data, PubMed, other countries' data, shortages	Parked in A005: re-weighting, a second AI problem, a second identity system, outside the three questions	Narrower evidence	A user need that one of them answers
Signals from the incremental FAERS path	Slower at two quarters (O2)	Full rebuild each time	A faster measurement at 4+ quarters
Spark for the clean layer	Polars on one machine is enough at this size	Not tested at scale	Data that outgrows one machine

How we know

Each row traces to a decision or cut: D009, D029, D031 v3 to v5 (v3 superseded; its cuts kept by v5), D043, A005 and the optimization log. The 'another month' plan (`docs/09-roadmap/another-month-sources-and-pharmacist-review.md`) puts a named pharmacist reviewer first, ahead of any new source.

Gate: Scope discipline

Required evidence: Every exclusion has a reason and a condition for revisiting it.

Result: Met for the twelve exclusions above.

If it fails: An exclusion with no reason looks like something we forgot.

09

Approaches we replaced or reverted

Every approach we try gets a file, including the ones that did not last. An honest note first: no approach file is marked 'Dropped' in the index today. The approaches below were replaced, reverted or never adopted, and their files are kept.

What we tried	What happened	Replaced by	Where
Opioids-only pilot class	Research scored a hybrid class higher	8 opioids + 5 GLP-1	A001 → A003, D014
v1 reference pack (30 cases, one source each)	Questions and answers written in one pass; capture errors	Reference set v2: frozen questions, independent verifier	A008, D037
Six-day build proposal	Not adopted; its ids (A006, D032 to D035) are reserved	12-step build order	A007, D036
One-page screen (question 1 only)	Did not answer recalls or changes	Four-page module	A012 → A016
Exact-term label matching alone	Synonym misses topped the GLP-1 lists	Three-step label check	A013 → A014, D043
Person-key duplicate fold alone	Missed one patient re-filed with changing ages	D023 v2 mass-report cut	A012
Incremental FAERS as the signals input (O2)	Correct but slower	Full rebuild stays the input	A015, D044
Column pruning in FAERS parsing (O3a)	Faster parse, no visible gain overall; reverted	Nothing (not the bottleneck)	Optimization log
A separate monitoring design (an unnumbered 'A00x' placeholder row in the index)	Parked as 'later'	A017 monitoring	INDEX.md

How we know

O2 and O3a are in the optimization log with their numbers, under 'Helped? No'. Keeping them is the point: the brief asks for what did not help, and a reviewer can check that each claim of 'faster' is backed by a measurement.

Gate: Record of what did not work TESTED

Required evidence: Replaced approaches keep their files, with the reason they were replaced.

Result: Met. Gap: the index uses 'Kept', 'Trying' and 'Replaces' instead of a 'Dropped' status, so the replaced approaches are spread across rows; a cleanup would mark A001 and the v1 pack as Dropped.

If it fails: If a dropped idea comes back without its history, we repeat the mistake.

A

Reference

Evidence and status labels

Label	Meaning
OBSERVED	Read directly from a public source (FAERS, DailyMed, openFDA, RxNorm), with the query that produced it.
DERIVED	Computed by our pipeline from observed data, for example a PRR or a count per drug and reaction.
ASSUMED	Not verified: taken as true for now because the data or the brief is silent. It could be wrong.
DESIGNED	A design choice we made and wrote down (the decision log says why). It may be built; the status band says how far.
TESTED	Built, run, and checked against evidence.
PLANNED	On the plan; not built yet.
PROPOSED	Suggested; waiting for a decision.

Figures quoted from project records (not in numbers.json)

These come from project records, not from the baseline-gate capture. Each is also cited where it appears.

Figure	Source
Morphine substring match about +50% rows	DQ34 (2026Q2 file, profiled 2026-10-20)
Primary-suspect-only drops more than half of oxycodone cases	DECISIONS learnings, 2026-11-04
Mass reports: 30 or more drugs per report; share 0.5 or more is not ranked	D023 v2
Foundry counts equal the 20 October prototype; five signals re-counted by hand	Learning 2026-11-04; baseline-gate.md
Ozempic label: openFDA copy months behind DailyMed	DQ28
F1 answer key: 35 items, Boxed Warning + Contraindications of 2 labels	eval/ai-eval-v1.md
DailyMed index about 30 pages per build; 'before' from the step 9 build	Optimization log O3
Drift needs a 3 to 4 week window	D009
Partitioning trigger: 8+ FAERS quarters or a build over 5 minutes	D048
Batch acceptance of 12 decisions; D041 v2 accepted (6 Nov 2026)	DECISIONS.md, decision changes
AI found diabetic-retinopathy terms, no optic-nerve term (semaglutide)	A019; eval/ai-eval-v1.md
FAERS replaced by AEMS on 11 March 2026	DQ20
Hybrid class scored higher than opioids only	D014; A003

Numbers used in this report

Every number in the table below is looked up from `docs/11-handin/numbers.json` at build time. Source: Foundry SQL (repo SQL panel) after the baseline-gate rebuild of the whole downstream chain: raw_dailymed_spl → clean_labels → clean_label_product_map → ontology_objects → ai_label_extraction → ontology_signals → ontology_changes → monitoring. Build durations from the Foundry Builds page. Query: docs/11-handin/baseline-gate.md. Captured: 2026-11-05T07:50Z. Figures quoted from other project records (decision log, data-issues register, optimization log) name that record where they appear.

Key	Value
ai.d015_target_f1	0.85
ai.d041_v2.model_calls	12
ai.gold_f1	0.97
ai.gold_recall_strict	35/35
ai.grounded	745
ai.model_calls_on_rebuild	0
ai.pilot_labels	5
ai.reactions	745
ai.sections	21
ai.v1_pilot.labels	2
changes.events_first_build	2,898
changes.last_30_days	26
changes.snapshot_diff_after_rebuild	0
faers.case_reactions	2,554,017
faers.cases_current	792,346
faers.drug_reaction_rows	373,522
faers.q1_cases	397,224
faers.q2_cases_deleted	655
faers.q2_cases_replaced	26,681
labels.total	1,321
monitoring.checks_per_build	21
monitoring.data_health_checks	4
monitoring.faers_age_days	130
monitoring.recalls_last4w	10
monitoring.recalls_prev4w	3
monitoring.reporter_type_blank_q1	0.343

Key	Value
monitoring.reporter_type_blank_q2	0.721
optimization.O1.after_calls	0
optimization.O1.after_tokens	0
optimization.O1.before_calls	9
optimization.O1.before_tokens	70,776
optimization.O2.full_rebuild_s	32
optimization.O2.incremental_add_q2_s	88
optimization.O3.after_downloads	2
optimization.O3.after_s	83
optimization.O3.before_downloads	1,321
optimization.O3.before_s	203
scope.drugs_formulary	13
scope.glp1	5
scope.opioids	8
signals.ae_stats	14,921
signals.label_check_stats.ai_term_exact	188
signals.label_check_stats.label_text	1,252
signals.label_check_stats.none	13,108
signals.label_check_stats.synonym_high	163
signals.label_check_stats.synonym_medium	210
signals.not_in_label	1,844
signals.possibly_in_label	108
signals.ranked	1,073
signals.ranked_history.d016_v3	1,115
signals.ranked_history.step9	1,355
signals.ranked_history.syn_v1	1,210
signals.ranked_history.syn_v2	1,073
signals.total	1,952
synonyms.high	69
synonyms.medium	72
synonyms.rejected	117

Key	Value
synonyms.rows	258
synonyms.version	syn_v2